



HelaDepDet: A Novel Multi-class Classification Model for Detecting the Severity of Human Depression

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Abstract. Depression-driven suicide is a serious social problem. Early identification of depression is vital for the well-being of society. Clinical diagnosis of depression takes a significant amount of time and requires highly skilled medical staff, which greatly limits its accessibility. Social media analysis for depression detection is therefore a rapidly growing research area. However, most of the available methods can only detect the presence or absence of depression, not the severity of depression. On the other hand, a few recently developed models for depression severity detection have not been validated on large datasets due to fundamental issues such as data sparsity. In this study, we proposed a novel method based on confidence vectors for detecting the severity of depression. We evaluated our method using a large dataset consisting of more than 40,000 annotated statements extracted from multiple social network services. To our knowledge, this is the largest and most well-balanced dataset for depression severity classification to date. Preliminary results showed that our models outperformed the existing state-of-the-art models by 5%, achieving a micro-averaged F1 score of 66% for human depression severity detection.

Keywords: Depression Severity Detection · Multi-class Classification · Natural Language Processing · Affective Computing

1 Introduction

Human depression has been identified as a severe mental illness throughout the entire world. According to the latest statistics of the World Health Organization, nearly 350 million people suffer from this mental disorder while over 700,000 people die due to suicide as a result of depression [1, 2]. Even though sufficient attention has been given to physical illnesses such as cancer, diabetes, allergies, etc., still most of the mental disorders such as schizophrenia, anxiety and depression have been neglected [3]. Among these, depression has been categorized as the most critical type of mental disorder globally [4]. The reason behind this situation is that more than 70% of depressed people do not seek early treatment due to various circumstances [5]. Even though many clinical approaches were established for diagnosing depression, the early identification of the severity of human depression remains a challenging task [6].

At the same time, the response to COVID-19 has played a major role in reshaping human society and working culture. Significant research has revealed that the major symptoms of human anxiety and depression can be elevated with sudden changes to our working culture [7]. Even though working from home has been one of the major impacts of the COVID-19 pandemic, serious mental health consequences have been reported due to the lack of social relationships and communication [8]. Evidence for this has been seen in the extraordinary increase in social media use in response to isolation from loved ones [9]. Considering all these factors, an early depression identification screening tool can be considered as a critical social need.

Multiple research challenges can be determined such as: “*How can the early identification of human depression be further enhanced?*” and, “*How can the categorization of the severity of human depression be further improved?*” In this study we evaluated a novel method for detecting depression severity based on social media analysis. The main contributions of this study can be summarized as follows:

- We proposed a novel method for human depression severity detection. Severity is formulated as a confidence vector representation of depression mentions such as words and/or phrases.
- We created a balanced dataset, which has been properly annotated according to the guidelines of the depression severity annotation scheme [10] and BDI-3 [11] clinical levels.
- We validated our method on the largest dataset consisting of more than 40,000 annotated statements extracted from multiple social media platforms. Our model was compared with state-of-art baselines for depression detection under several benchmarking strategies to show that the proposed methodology performs well at depression severity detection.

The overall organization of the paper is as follows: Sect. 2 critically reviews related work in this area while Sect. 3 explains the data construction and the overall methodology. Experimental results are presented in Sect. 4 and then discussed in Sect. 5.

2 Related Work

The literature can be grouped into two main segments which are analyzing human depression detection and analyzing the severity detection of human depression. Although human depression detection approaches can be identified as clinical, emotional, or linguistic based, this research work has focused only on linguistic based depression detection. When considering any text based linguistic approach, a significant public corpus is essential. Losada, et al., [12] were the first to publish a corpus related to human depression detection. A further enhancement was made by revealing the nature of human depression and language usage as the first in a series of eRisk challenges [13]. Shen, et al., [14] were able to extract depression related feature groups on Twitter. After 2017, a new trend arose where human depression detection benefitted from natural language processing (NLP) and a novel approach has been proposed for extracting the linguistic relationships for depression detection [15]. Skaik, et al., [16] presented a systematic review on the significant progress of human depression identification based on social media.

Suhara, et al., [17] proposed a machine learning driven approach for forecasting human depressive moods using logs extracted from personal devices as inputs. Orabi, et al., [18] demonstrated a convolutional neural network (CNN) based approach for obtaining features of language that are useful in depression detection. The first attempt has been made to use transfer learning to implement a computational model for human depression identification [19]. Burdisso, et al., [20] introduced a novel binary classification machine learning model to detect depression identification using social media postings. A specific study on depression detection using text summarization has been released as a deep neural based framework [21].

Since severity detection of human depression is the focus of our approach, investigating such attempts is important. Even though there is a long history of detecting the severity of human depression based on clinical or emotional procedures, little work has been conducted using text based linguistic methods. An ensemble machine learning approach has been introduced for depression severity detection where a Bidirectional Encoder Representations from Transformers (BERT) model has been applied as a pre-trained language model (PLM) [22]. Wu, et al., [23] demonstrated an enhanced version of the previous attempt for measuring the early signs of depressive severity disorders. In the same year, a constructive survey has been conducted on listing the trending NLP approaches in depression severity identification [24]. Recently, Naseem, et al., [25] introduced a deep neural based approach for depression severity classification using social media posts. Although different approaches have been devised, major issues remain to be addressed such as the shortage of text based linguistic public corpora and unbalanced data distribution issues in the prevailing corpus.

3 Methodology

We can divide our methodology into two phases, data construction and algorithmic development.

3.1 Data Construction

Depression severity annotation is diversified considering the clinical depression categorization. As per the text based linguistic approach, the depression severity annotation has been conducted considering BDI-3 [11] and the depression severity annotation scheme of Mowery, et al., [10] as the ground truth. According to the criteria, depression can be categorized into four levels: *minimal*, *mild*, *moderate*, and *severe*. Two such publicly available annotated corpora have been considered, namely DEPTWEET [26] and the depression severity annotated dataset of Naseem, et al. [25]. These datasets were developed using the extracted social media posts from Twitter and Reddit users, although both suffer from structural issues such as insufficient size and unbalanced data distribution among the pre-defined classes. To overcome these issues, an aggregated larger corpus has been introduced as a part of the present work. The NearMiss [27] under-sampling technique has been used to balance data tuples optimally. Common features of both datasets such as *id*, *reply_count*, *reply_comment*, *reply_addressee*, *target_label*,

Table 1. Class-wise corpora distribution.

Dataset	Depression class	Corpus size
Depression severity annotated corpus	Minimal	2,587
	Mild	290
	Moderate	394
	Severe	282
DEPTWEET	Minimal	32,401
	Mild	5,243
	Moderate	1,810
	Severe	741
Our aggregated corpus	Minimal	10,549
	Mild	10,661
	Moderate	9,473
	Severe	11,176

and *confidence_score* were used for corpora aggregation. The class-wise distribution statistics of the corpora are shown in Table 1.

It can be seen that the tuple distribution of the aggregated corpus is well balanced after applying the NearMiss undersampling strategy which is one significant contribution of our study. A sample of the constructed aggregated corpus is shown in Fig. 1.

<i>"I am gaming with my cousin every evening but he is making fun of me and trash telling me recently more and more Especially when others are around."</i>	Minimal
<i>"My mind frame is so low. I feel like I just do not even want to be here anymore."</i>	Mild
<i>"I am so scared of everything, when will it stop? scared"</i>	Moderate
<i>"I feel so empty. I feel like I am a machine. I am sick of repeating the same thing every day."</i>	Severe

Fig. 1. Sample of the aggregated constructed dataset

The labelling procedure has been conducted considering the highlighted depression related terms. The original dataset labels were used as the annotations of the aggregated

dataset so that combining them would not have any impact on the fundamentals of corpus construction.

3.2 Methodological Development

A high-level conceptual overview of our methodology is shown in Fig. 2.

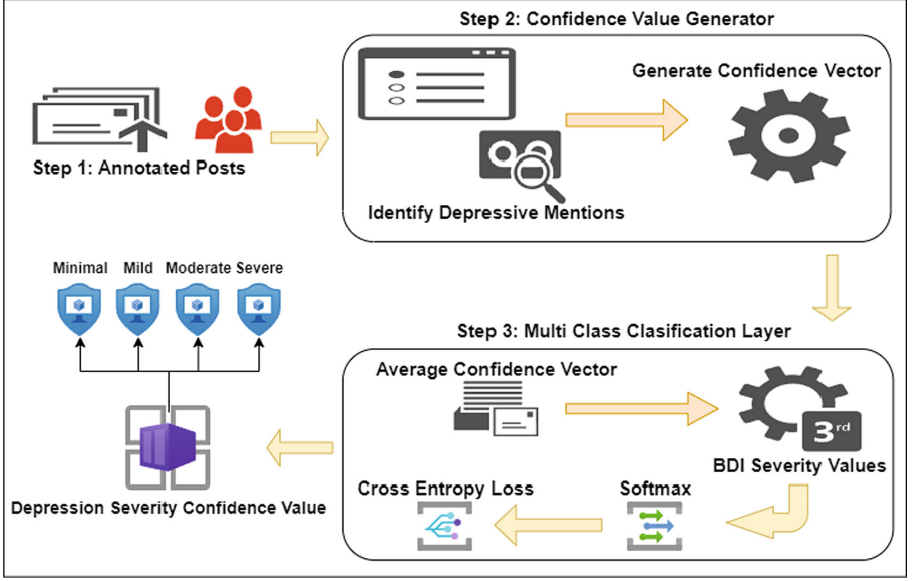


Fig. 2. The high-level architecture of our method

The entire approach can be divided into three main steps. In the initial step, the annotated social media posts were fed to the system. The pre-annotation tasks such as preprocessing are performed in this step. Identification of the depressive words/phrases and generating the confidence vectors are the tasks of step 2: *confidence value generator*. Identifying depressive mentions is facilitated by prior work on extracting hate speech related mentions [28]. The respective depression severity categories (*minimal*, *mild*, *moderate*, and *severe*) g and the potential depressive mentions (*words* and/or *phrases*) m are mapped to form a dictionary. Considering these, the depression confidence vector cv of a particular social media conversational post can be obtained as in Eq. 1.

$$cv = \{cv(m_1g_1), cv(m_2g_2), cv(m_3g_3), \dots, cv(m_ng_n)\} \quad (1)$$

Generating the confidence vectors for the determined depressive words/phrases is another novel contribution of our work. Once the confidence vector for each user post is obtained, the third section *multi-class classification layer* commences. This section is responsible for deriving the multi-class depression severity classification. The average

confidence vector CV can be determined as in Eq. 2.

$$CV = \sum_{mg \in t_s} cv(m_n g_n) \quad (2)$$

The set t_s denotes the conversation history of a particular user. In a text-based multi-party conversation, conversation history (or interaction history) is essential for the semantic mapping of each follow-up utterance of the same conversation to identify the transition of conversation topics and for dynamic topic tracking [29]. This mapping is done using the above-mentioned *reply_count* and *reply_comment* features. Once the CV is obtained, it is converted to the corresponding BDI-3 score for determining the level of depression severity. A novel approach to multi-class depression severity classification was used that involved several steps including sentence classification, n-gram mapping, classification of n-grams, and layer normalization. The sentence classification layer determines classified utterances along with labels of depression severity levels. These classified utterances are tokenized for obtaining n-grams. Once n-grams are obtained, n-gram mapping with the respective CV is determined using a direct mapping approach [30]. This mapping layer consists of a fully connected deep network with a hidden layer which contains 12 hidden neurons. The classification of n-grams is shown in Algorithm 1 (Fig. 3).

ALGORITHM 1: Classification of n-grams

```

Input -> self, x_test, prep, leave_pbar
Output -> cv: a list of numeric values representing the confidence levels for each element in x_test

If self.__categories__ is empty, raise an EmptyModelError
If self.get_ngrams_length() and self.get_nword_slices() is true
then
    return self.__predict_fast__(x_test, prep=prep,
    leave_pbar=leave_pbar, proba=True) End If
Set x_test to be a list of x_test
Set classify to be self.classify()
for each element x in x_test
    Return a list of classifications cv
    Iterate over each element x in x_test
END FOR
RETURN cv

```

Fig. 3. Classification of n-grams

N-gram classification is conducted using N-word slices. This helps to obtain BDI-3 scores for respective depressive words and/or phrases which are generated as in Fig. 2 by converting the determined CV into the respective BDI-3 score. A softmax activation function and a normalization layer enhance this process while normalizing mappings. This enables smoother gradients, a faster training process, and ultimately better accuracy

[31]. Cross entropy [32] is used as the loss function for minimizing the loss between the generated and predicted results of the system. The loss function LV can be determined as:

$$LV = - \sum_{g=1}^M y_{mg} \log(p_g) \quad (3)$$

where $M = |\{Minimal, Mild, Moderate, Severe\}|$ is the number of depressive severity levels, y is a binary value indicating whether g is the correct classification and p is the probability of an exact depressive category. Assuming the respective severity level is s , the depression severity confidence value $DSCV$ can be obtained as in Eq. 4.

$$DSCV = \left[softmax(CV\{s, M\}), - \sum_{g=1}^M y_{mg} \log(p_g) \right] \quad (4)$$

The obtained $DSCV$ lies in the range 0 to 63, aligning with the range of the BDI-3 scale. Considering the range of $DSCV$, the depression severity levels can be determined as: *minimal* (0–9), *mild* (10–18), *moderate* (19–29) or *severe* (30–63), aligning with the levels of BDI-3 scale.

For the implementation, *Python 3.8* was used for core model development and *TensorFlow 2.10* was used as the backend machine learning framework. Several NLP related Python packages such as *NLTK* were used to support various pre-processing related tasks. The model training was performed for 10 epochs using a Nvidia GeForce RTX 3060 Ti GPU by setting the batch size to 4.

4 Experimental Results

Experiments on inter-annotator reliability and depression severity confidence value are discussed in this section.

4.1 Inter-annotator Reliability

Since multiple raters and multiple categorical ratings were employed in the construction of the original two corpora, the annotator’s reliability must be determined. The objective is not to re-annotate but to re-evaluate the labelled corpora after corpora aggregation. A sample of the annotated aggregated corpus is shown in Table 2.

Annotators 1, 2 and 3 represent the panel of annotators while Trust 1, 2 and 3 are confidence coefficients on a scale of 0 to 1 for the obtained annotator values. Annotator values range from 1 to 4 representing the depression severity classes from *minimal*, to *severe*, respectively. The most appropriate depression severity class has been determined considering the maximum number of votes from the panel of annotators. The respective severity labels and confidence scores for social media statements can be found as follows in Table 3.

The inter-annotator agreement was determined by calculating the Kappa coefficient (K) [33] which measures the pairwise agreement among a set of annotators to obtain a

Table 2. Samples of the aggregated corpus.

Post	Annotator 1	Annotator 2	Annotator 3	Trust 1	Trust 2	Trust 3
<i>He said he had not felt that way before, suggested I go rest...</i>	1	1	2	0.967	0.967	1
<i>I decided to stop taking it for a few days to see if it was really helping. Now, about 40 h...</i>	3	2	3	1	1	0.933

Table 3. Samples of confidence scores and target labels.

Post	Target label	Confidence score
<i>He said he had not felt that way before, suggested I go rest...</i>	Minimal	0.647
<i>I decided to stop taking it for a few days to see if it was really helping. Now, about 40 h since my last dose...</i>	Moderate	0.667

reliability measure for a manual labelling task under a computational task. The above confidence value was used as an average agreement of all annotators. Considering these, the K coefficient is determined as follows.

$$K = \frac{\text{observed agreement} - \text{chance agreement}}{1 - \text{chance agreement}} \quad (5)$$

The inter-annotator reliability can be classified as a *fair*, *moderate*, *substantial*, or *almost perfect* for values of K in the ranges 0.21 - 0.40, 0.41 - 0.60, 0.61 - 0.80 or 0.81–0.99 respectively [34]. The observed agreement indicates the proportion of times that the annotators agreed where chance agreement is considered as the maximum proportion of times that annotators could agree. In our case, the obtained confidence score can be taken as the observed agreement. The individual Kappa values K for the four depression severity categories are shown in Table 4.

Table 4. Kappa values for depression severity categories.

Class	Probability	Kappa	Error	Z value	P value
Minimal	0.82	0.75	0.13	3.42	0.04
Mild	0.77	0.68	0.13	3.86	0.04
Moderate	0.69	0.72	0.13	4.21	0.03
Severe	0.57	0.71	0.13	3.58	0.03

Here, *probability* is the chance of being annotated by all annotators which is important for obtaining the probability of chance agreement. *Error* is for the standard error of the calculated *K* which measures the overall precision of the estimation. The *Z value* represents an approximate normal testing statistic, and the *P value* is the probability of measuring evidence against the null hypothesis. Considering the obtained individual Kappa and *P* values, we can conclude that all individual class level inter-annotator reliability shows a substantial good agreement with statistical significance.

4.2 Depression Severity Confidence Value

The next analysis examines the performance of our models on *DSCV*. The model was evaluated using three performance measures: *Precision*, *Recall*, and the *F1 Score*. These measures were calculated using the following equations where *TP*, *FP*, *TN*, *FN* represent true positives, false positives, true negatives, and false negatives respectively. Additionally, the micro-average Precision, micro-average Recall and micro-average F1 score were also calculated from the confusion matrix shown in Table 5.

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

Table 5. Confusion matrix.

	Annotated as depressed	Annotated as non-depressed
Are recognized as depressed	True Positives	False Negatives
Are recognized as non-depressed	False Positives	True Negatives

We used a one-versus-rest strategy where the positive class refers to one of the depression levels and the negative class refers to all the other levels. The set of evaluation metrics were calculated for each of the four cases where each depression level was considered as the positive class, and the average values of the metrics were used as the results.

Ten-fold cross validation was used to avoid overfitting and selection bias. The grid search optimization technique [35] was used to optimize the hyper-parameters. The warm-up proportion, learning rate, epoch rate, and batch size were set to 0.1, 0.005, 10, and 4 respectively. The obtained experimental results for *DSCV* are shown in Table 6. *P*, *R*, and *F1* stand for Precision, Recall, and F1 Score respectively.

The micro-averaged values of the above statistical experimental results were obtained since *micro-averaged Precision*, *micro-averaged Recall*, and *micro-averaged F1 Score*

Table 6. Class-wise experimental results for each fold of the cross-validation process.

		k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8	k=9	k=10
Minimal	<i>P</i>	0.692	0.693	0.693	0.696	0.698	0.703	0.712	0.718	0.722	0.724
	<i>R</i>	0.676	0.677	0.678	0.677	0.678	0.678	0.679	0.681	0.681	0.682
	<i>F1</i>	0.688	0.688	0.689	0.689	0.689	0.691	0.691	0.692	0.692	0.693
Mild	<i>P</i>	0.686	0.686	0.682	0.683	0.685	0.687	0.687	0.688	0.689	0.691
	<i>R</i>	0.627	0.627	0.628	0.628	0.628	0.629	0.629	0.63	0.631	0.632
	<i>F1</i>	0.643	0.643	0.645	0.646	0.647	0.647	0.648	0.649	0.649	0.651
Moderate	<i>P</i>	0.663	0.664	0.665	0.665	0.666	0.667	0.667	0.668	0.668	0.669
	<i>R</i>	0.602	0.603	0.604	0.604	0.605	0.605	0.605	0.606	0.607	0.607
	<i>F1</i>	0.576	0.577	0.577	0.578	0.579	0.579	0.581	0.582	0.582	0.583
Severe	<i>P</i>	0.716	0.716	0.717	0.717	0.718	0.719	0.719	0.72	0.721	0.722
	<i>R</i>	0.637	0.637	0.638	0.638	0.639	0.639	0.64	0.641	0.641	0.642
	<i>F1</i>	0.655	0.655	0.656	0.657	0.657	0.658	0.659	0.659	0.661	0.662

Table 7. Average results for DSCV generation.

Class	Precision	Recall	F1 Score
Minimal	0.71	0.68	0.69
Mild	0.69	0.63	0.65
Moderate	0.67	0.61	0.58
Severe	0.72	0.64	0.66
Micro Avg	0.68	0.65	0.66

have been considered as the optimal evaluation metrics for evaluating multi-class classification models [10]. The class-wise micro-averaged results shown in Table 7.

These experimental results were used for benchmarking purposes. Our results were compared with the results of the existing state-of-the-art baselines for text-based depression severity detection. These baselines have been introduced for addressing downstream tasks such as depression detection and self-harming detection using text-based corpora. These baselines were set up locally and we conducted several evaluation cycles using our novel aggregated corpus to satisfy the fundamental requirements of benchmarking. Regarding baselines, *DEPTWEET* [26] and *DepressionNet* [21] were used as depression classification machine learning models. *SVM* + *LBFs*, *RF* + *LBFs* and *MLP* +

LBFs were used as self-harming detection models. *LBFs* stands for language-based features such as term frequency-inverse document frequency (*TF-IDF*), parts of speech, etc. Such features were used as inputs to different machine learning classifiers such as support vector machines (*SVM*), random forest (*RF*), and multi-layer perceptron (*MLP*). The benchmarking results are shown in Table 8.

Table 8. Comparison to prior studies.

Model	Precision	Recall	F1 Score
DepressionNet	0.61	0.59	0.60
DEPTWEET	0.62	0.61	0.61
SVM + LBFs	0.51	0.56	0.52
RF + LBFs	0.59	0.54	0.56
MLP + LBFs	0.56	0.54	0.56
Ours - <i>HelaDepDet</i>	0.68	0.65	0.66

Considering the benchmarking results, we can conclude that our novel model performed better than existing baselines in *micro-average Precision*, *micro-average Recall*, and *micro-average F1 Score*. The benchmarking was conducted considering only publicly available baselines for depression detection tasks since some baselines are private and their implementation cannot be accessed.

To summarize the benchmarking results, the hand-crafted feature-centric models such as *SVM + LBFs*, *RF + LBFs*, and *MLP + LBFs* performed slightly worse than the linguistic feature-based models such as *DepressionNet*, *DEPTWEET*, and our novel model -- *HelaDepDet*. The linguistic feature-based models performed well due to their ability to capture and categorize mental health state efficiently. Considering *DepressionNet*, *DEPTWEET*, and our novel model *HelaDepDet*, *HelaDepDet* performed well at capturing the exact depression severity level considering the overall loss of multi-class classification which is relevant to how exactly the predicted depression severity level differs from the real depression severity level.

Additionally, measuring the accuracy of our *DSCV* model for training the aggregated corpus is necessary. Model accuracy is one of the most popular metrics in multi-class classification and it is directly generated using the confusion matrix [36]. The accuracy indicates an overall evaluation of the quality of the model’s predictions considering the entire corpus. The evaluation metric for accuracy can be calculated as follows.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

The accuracy values of the overall model training process are shown in Fig. 4.

We can conclude that the accuracy increased with the number of epoch cycles of the training process. The maximum accuracy obtained after 10 epoch cycles was 0.678.

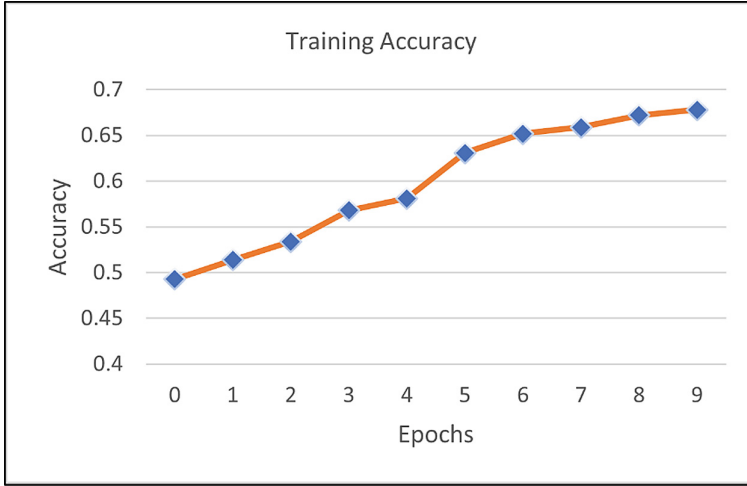


Fig. 4. Training accuracy

5 Discussion

Depression severity determination has been identified as essential for the well-being of human beings. With the impact of recent circumstances, the number of reported cases of human depression has increased at an alarming rate. Even though plenty of clinical depression detection practices are established, a significant number of depression-driven suicides are still reported worldwide. This reminds us of the high demand for an early identification depression detection tool. Since humans are social creatures, cases of human depression can be identified from recent behavioral changes more effectively than by traditional clinical depression detection [37]. In this study, we suggest a novel, multi-class classification approach to linguistic text-based depression severity detection, which can be used as an early depression detection screening tool based on social media analytics. Even though human depression can be analyzed using clinical, emotional, and linguistic techniques, we have concentrated on linguistic text-based depression severity detection in this study.

The initial contribution of our work is a novel aggregated dataset¹ for depression severity detection using BDI-3 [11] and the depression severity annotation scheme of Mowery, et al., [10] as the ground truth. Multiple depression severity annotated corpora were used for the novel dataset construction. Our aggregated corpus consists of more than 40,000 user posts and comments which were extracted from multiple social media networks including Twitter and Reddit. Four levels of clinical depression severity, *minimal*, *mild*, *moderate*, and *severe*, were used as labels to annotate the contents of the original datasets that we considered. Introducing this novel aggregated corpus helped to mitigate the existing issues with the available annotated corpora such as data sparsity and unbalanced data distribution among depression severity classes. Optimal undersampling techniques were used to balance data tuples. The depression severity annotations of the

¹ https://github.com/CyraxSector/Depression_Severity_Levels_Dataset.

original corpora were used as the annotations of the aggregated dataset so that combining them would not have any impact on the fundamentals of the corpus. Our novel corpus has been contributed back to the original authors and is stored in a public repository for research related purposes.

To address the two main research questions, “*How can the early identification of human depression be further enhanced?*” and “*How can the categorization of the severity of human depression be further improved?*”, we introduced a novel multi-class classification approach for human depression severity detection. Here, *early identification of human depression* can be identified as the most important result when using our novel model to classify text-based chat conversations for human behavioral analysis. On social media platforms there is a high tendency for users to express feelings and emotions without hiding anything and therefore analyzing that content could lead to the detection of abnormal behavior sooner than would analyzing the same cohort clinically at a later stage. We formulated depression severity as a confidence vector considering depression related words and/or phrases. To map the generated confidence vector to the corresponding BDI-3 score, several steps such as sentence classification, n-gram mapping, classification of n-grams, and layer normalization were used. The process as a whole can be recognized as a novel approach to multi-class depression severity classification and the research question, “*How can the early identification of human depression be further enhanced?*” can be considered answered. Our novel model was implemented² using *Python 3.8* as the main programming language.

Our model was evaluated using multiple experimental metrics. The main novelty, which is obtaining the depression severity confidence value *DSCV*, was evaluated according to *Precision*, *Recall*, *F1 score*, and *accuracy*. From the evaluation results, we conclude that our model outperformed the existing state-of-the-art models for depression severity detection as a multi-class classification task by a considerable margin. Considering our benchmark results we can say that the next research question, “*How can the categorization of the severity of human depression be further improved?*”, has also been addressed.

Regarding ethical concerns, since the annotated tuples of original corpora which were used for aggregation purposes were already published, an additional ethical approval is not needed. However, to avoid the risk of information leakage, all extracted social media posts were anonymized. The results of inter-annotator reliability re-evaluations will be made available in the near future with a corresponding non-disclosure agreement.

6 Limitations and Future Work

Considering the sustainability of corpora aggregation, several issues may occur such as biases due to multiple annotators and demographic or other social factors. Most of these issues have been mitigated by re-evaluating the inter-annotator reliability of the aggregated corpora following a constructive evaluation process. We are working on mitigating other issues and limitations as far as possible before this study is converted into a finished system. Although we conducted experiments on a large and well-balanced dataset,

² <https://github.com/CyraxSector/HelaDepDet>. The code will be released publicly in the near future for research purposes.

the experiments were limited to depression classification machine learning models and hand-crafted feature-centric models. In future, we hope to conduct further experiments on larger pre-trained language models for constructive benchmarking purposes. Additionally, this research is limited to a screening tool for depression severity detection that must not be used for clinical diagnosis purposes.

In future we will focus on analyzing multi-party conversations in social media and investigate how multi-party conversations can be used to determine the severity of human depression. Additionally, experiments with clinical and emotional based approaches in multi-model human depression detection would be valuable future work.

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References

1. Hsu, C.W., Tseng, W.T., Wang, L.J., Yang, Y.H., Kao, H.Y., Lin, P.Y.: Comparative effectiveness of antidepressants on geriatric depression: real-world evidence from a population-based study. *J. Affect Disord.* **296**, 609–615, 1 January 2022. <https://doi.org/10.1016/j.jad.2021.10.009>. Epub 14 October 2021 PMID: 34655698 (2022)
2. Kaplan, C., Zhang, Y.: Assessing the comparative-effectiveness of antidepressants commonly prescribed for depression in the US Medicare population. *J. Ment. Health Policy Econ.* 171–178. PMID: 23525835; PMCID: PMC3608926 (2012)
3. Wang, D., Liu, L.: The depression of Chinese and the reflection related to their society and culture. *Chin. General Pract.* **7**(5), 315–317 (2004)
4. Zafar, A., Chitnis, S.: Survey of depression detection using social networking sites via data mining. In: 10th International Conference on Cloud Computing, Data Science and Engineering (Confluence), pp. 88–93 (2020)
5. Shen, G., et al.: Depression detection via Harvesting social media: a multimodal dictionary learning solution. In: International Joint Conference on Artificial Intelligence (2017)
6. Rissola, E.A., Aliannejadi, M., Crestani, F.: Beyond modelling: understanding mental disorders in online social media. In: Jose, J., et al. *Advances in Information Retrieval. ECIR 2020. Lecture Notes in Computer Science*, vol. 12035. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-45439-5_20
7. Li, H.Y., Cao, H., Leung, D.Y., Mak, Y.W.: The psychological impacts of a COVID-19 outbreak on college students in China: a longitudinal study. *Int. J. Environ. Res. Public Health* **17**(11), 3933 (2020)
8. Gruber, J., et al.: Mental health, and clinical psychological science in the time of COVID-19: challenges, opportunities, and a call to action. *Am. Psychol.* (2020). Advance online publication. <https://doi.org/10.1037/amp0000707>
9. Karhu, M., Suoheimo, M., Häkkinä, J.: People's perspectives on social media use during COVID-19 pandemic. In: 20th International Conference on Mobile and Ubiquitous Multimedia (MUM 2021). Association for Computing Machinery, New York, NY, USA, pp. 123–130 (2022)
10. Mowery, D.L., Bryan, C., Conway, M.: Towards developing an annotation scheme for depressive disorder symptoms: a preliminary study using twitter data. In: *Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology: From Linguistic Signal to Clinical Reality*, pp. 89–98 (2015)

11. Jackson-Koku, G.: Beck depression inventory. *Occup. Med.* **66**(2), 174–175 (2016)
12. Losada, D.E., Crestani, F.: A test collection for research on depression and language use. In: Fuhr, N., et al. (eds.) *CLEF 2016*. LNCS, vol. 9822, pp. 28–39. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-44564-9_3
13. Losada, D.E., Crestani, F., Parapar, J.: eRISK 2017: CLEF lab on early risk prediction on the internet: experimental foundations. In: Jones, G.J.F., et al. (eds.) *CLEF 2017*. LNCS, vol. 10456, pp. 346–360. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-65813-1_30
14. Shen, G., et al.: Depression detection via harvesting social media: a multimodal dictionary learning solution. In: *International Joint Conference on Artificial Intelligence*, pp. 3838–3844 (2017)
15. Burdisso, S., Errecalde, M.L., Montes y Gómez, M.: To-wards measuring the severity of depression in social media via text classification. In: *XXV Congreso Argentino de Ciencias de la Computación (CACIC)* (Universidad Nacional de Río Cuarto, Córdoba (2019)
16. Skaik, R., Inkpen, D.: Using social media for mental health surveillance: a review. *ACM Comput. Surv. (CSUR)* **53**(6), 1–31 (2020)
17. Suhara, Y., Xu, Y., Pentland, A.S.: DeepMood: forecasting depressed mood based on self-reported histories via recurrent neural networks. In: *Proceedings of the 26th International Conference on World Wide Web*, pp. 715–724 (2017)
18. Orabi, A.H., Buddhitha, P., Orabi, M.H., Inkpen, D.: Deep learning for depression detection of twitter users. In: *Proceedings of the Fifth Workshop on Computational Linguistics and Clinical Psychology: From Keyboard to Clinic*, pp. 88–97 (2018)
19. Abed-Esfahani, P., et al.: Transfer learning for depression: early detection and severity prediction from social media postings. *CLEF (working notes)* **1**, 1–6 (2019)
20. Burdisso, S.G., Errecalde, M.L., Montes y Gómez, M.: Using text classification to estimate the depression level of Reddit users. *J. Comput. Sci. Technol.* (2021)
21. Zogan, H., Razzak, I., Jameel, S., Xu, G.: DepressionNet: a novel summarization boosted deep framework for depression detection on social media. *arXiv preprint arXiv:2105.10878* (2021)
22. Rao, G., Peng, C., Zhang, L., Wang, X., Feng, Z.: A knowledge enhanced ensemble learning model for mental disorder detection on social media. In: Li, G., Shen, H.T., Yuan, Y., Wang, X., Liu, H., Zhao, X. (eds.) *KSEM 2020*. LNCS (LNAI), vol. 12275, pp. 181–192. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-55393-7_17
23. Wu, S.H., Qiu, Z.J.: A RoBERTa-based model on measuring the severity of the signs of depression. In: *CLEF (Working Notes)*, pp. 1071–1080 (2021)
24. Ríssola, E.A., Losada, D.E., Crestani, F.: A survey of computational methods for online mental state assessment on social media. *ACM Trans. Comput. Healthcare* **2**(2) (2021)
25. Naseem, U., Dunn, A.G., Kim, J., Khushi, M.: Early identification of depression severity levels on Reddit using ordinal classification. In: *Proceedings of the ACM Web Conference*, pp. 2563–2572 (2022)
26. Kabir, M., et al.: DEPTWEET: a typology for social media texts to detect depression severities. *Comput. Hum. Behav.* **139**, 107503 (2023)
27. Yang, Z., Gao, D.: An active under sampling approach for imbalanced data classification. In: *2012 Fifth International Symposium on Computational Intelligence and Design*, vol. 2, pp. 270–273. IEEE (2012)
28. Priyadarshana, Y.P., Ranathunga, L., Amalraj, C.R.J., Perera, I.: HelaNER: a novel approach for nested named entity boundary detection. In: *IEEE EUROCON 2021-19th International Conference on Smart Technologies*, pp. 119–124. IEEE (2021)
29. Gu, J.C., Tao, C., Ling, Z.H.: WHO Says WHAT to WHOM: a survey of multi-party conversations. In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI*, pp. 5486–5493 (2022)

30. Chen, S., Wang, J., Jiang, F., Lin, C.Y.: Improving entity linking by modeling latent entity type information. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, No. 05, pp. 7529–7537 (2020)
31. Zhu, Q., He, Z., Zhang, T., Cui, W.: Improving classification performance of softmax loss function based on scalable batch-normalization. *Appl. Sci.* **10**(8), 2950 (2020)
32. Gordon-Rodriguez, E., Loaiza-Ganem, G., Pleiss, G., Cunningham, J.P.: Uses and abuses of the cross-entropy loss: case studies in modern deep learning (2020)
33. Sim, J., Wright, C.C.: The Kappa Statistic in Reliability Studies: Use, Interpretation, and Sample Size Requirements. *Phys. Ther.* **85**(3), 257–268 (2005)
34. Chen, B., Zaebs, D., Seel, L.: A macro to calculate kappa statistics for categorizations by multiple raters. In: Proceeding of the 30th Annual SAS Users Group International Conference, pp. 155–130 (2005)
35. Pontes, F.J., Amorim, G.F., Balestrassi, P.P., Paiva, A.P., Ferreira, J.R.: Design of experiments and focused grid search for neural network parameter optimization. *Neurocomputing* **186**, 22–34 (2016)
36. Grandini, M., Bagli, E., Visani, G.: Metrics for multi-class classification: an overview. arXiv preprint [arXiv:2008.05756](https://arxiv.org/abs/2008.05756) (2020)
37. Guntuku, S.C., Yaden, D.B., Kern, M.L., Ungar, L.H., Eichstaedt, J.C.: Detecting depression and mental illness on social media: an integrative review. *Curr. Opin. Behav. Sci.* **18**, 43–49 (2017)